Alexander Cote

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Summary

Software engineer (B.S., Computer Science and Information Technology, RPI) building production full stack systems: a FastAPI backend on AWS Lambda and Aurora Postgres, React/TypeScript frontends, and Terraform-managed infrastructure. Builds AI systems including RAG assistants (pgvector, ChromaDB) and a LangGraph agentic coaching workflow. Background in security coursework and CTFs, with a focus on reliability, testing, and auditable system design.

Experience

Homefield Capital Partners | Troy, NY
Full Stack Developer | 01/2026 - Present

- Build backend services and a large REST API for a production investment platform used by fund managers and economic development programs (FastAPI, PostgreSQL, AWS Lambda).
- Develop financial features including investor commitments, e-signatures, a capital ledger, and tax reporting.
- Implement security and compliance features (MFA, role-based database access, audit logging, encrypted data storage) and contribute to CI/CD, testing, the React frontend, and Terraform infrastructure.
- Built a RAG assistant over a large knowledge base using vector search and OpenAI models, with responses tailored to different user types.
- Developed an agentic AI coaching workflow with LangGraph, including LLM-based evaluation and automated testing with simulated users.

Education

Rensselaer Polytechnic Institute | Troy, NY
B.S. Computer Science & Information Technology, GPA 3.3 | 05/2026

- Relevant Coursework: Database Systems, Hardware Security, Network Programming, Computer Organization, Information Systems Security, Web Systems Development

Skills

- **Languages:** Python, Java, TypeScript, JavaScript, SQL, C++, HTML/CSS, PHP
- **Backend & Frameworks:** FastAPI, SQLAlchemy, Alembic, REST APIs, React, Node.js, Vite
- **Cloud & DevOps:** AWS (Lambda, API Gateway, Aurora, SQS, S3, CloudFront, Cognito), Terraform, Docker, GitHub Actions CI/CD, Azure, Google Cloud
- **Data:** PostgreSQL, MongoDB
- **AI/ML:** RAG, agentic AI, MCP, LLM integration (OpenAI), vector embeddings (Sentence Transformers)

Projects

Employee Compliance Assistant, Deloitte Capstone Project (Fall 2025)
Lead Backend Developer · Python | FastAPI | LangChain | ChromaDB | React | MariaDB | Docker

- Built the backend for a compliance platform that tracks employees' financial institution holdings and answers policy questions through a RAG chat assistant (Sentence Transformers embeddings, ChromaDB, Llama 3.3 70B via LangChain) with citations to source document pages.
- Designed PII guardrails that mask emails, SSNs, card and account numbers, phone numbers, and personal names (regex plus spaCy named-entity recognition) in both user prompts and model responses before anything is stored or shown.
- Implemented JWT authentication, persistent chat sessions with citations, and annual compliance cycle logic, deployed as a multi-container Docker Compose stack.